

STATEMENT OF WORK

I. GENERAL INFORMATION

A. Introduction

The USGS Geology, Minerals, Energy, and Geophysics (GMEG) Science Center requires whole-rock samarium-neodymium, uranium-lead, and rubidium strontium isotope analyses of samples collected from mafic-ultramafic igneous (cumulates) and evolved intrusive igneous (porphyries) rocks. These isotope analyses will support ongoing research for critical mineral studies of rocks associated with arc magmatism, copper porphyry formation, and the mission of Earth MRI.

B. Background

In igneous rocks, isotopes systems are used to determine melt sources (e.g., mantle plume or sub-continental mantle lithosphere) and extent of contamination of those melts. The combination of Rb-Sr, U-Pb, and Sm-Nd isotope system has been widely used to study porphyry copper deposits as well as the origin of ultramafic cumulates. These data will help to understand the petrogenesis of porphyry deposits USGS Earth MRI focus areas, including the Wikieup and the Greyback plutons, AZ, as well as ultramafic xenoliths associated with mineralization in central AZ. The Rb-Sr, U-Pb, Sm-Nd isotopes will help to elucidate the sources setting(s) of these magmatic events and their degree of contribution.

C. Definitions/Applicable Documents

Briefly, the methods for separation and isotopic measurement of the parent and daughter nuclides involve first dissolving whole-rock sample powders with concentrated acid in pressurized (Parr) vessels. For the U-Pb and Sm-Nd systems the dissolutions are then spiked with a tracer solution. Parent and daughter nuclides are then isolated by cation exchange chromatography. Once purified, parent and daughter isotopic compositions are measured by multi-collector inductively coupled plasma mass-spectrometry (MC-ICP-MS).

Literature that provides an overview of the methods:

- Cheng, H., King, R. L., Nakamura, E. (2008). Coupled Lu–Hf and Sm–Nd geochronology constrains garnet growth in ultra-high-pressure eclogites from the Dabie orogen. *J. Metamorphic Geol.*, v. 26, p. 741–758.
- White, William M., Francis Albarède, and Philippe Télouk. “High-Precision Analysis of Pb Isotope Ratios by Multi-Collector ICP-MS.” *Chemical Geology* 167, no. 3 (2000): 257–70. [https://doi.org/10.1016/S0009-2541\(99\)00182-5](https://doi.org/10.1016/S0009-2541(99)00182-5).
- Gaschnig, Richard M., Jeffrey D. Vervoort, Reed S. Lewis, and Basil Tikoff. “Isotopic Evolution of the Idaho Batholith and Challis Intrusive Province, Northern US Cordillera.” *Journal of Petrology* 52, no. 12 (2011): 2397–429. <https://doi.org/10.1093/petrology/egr050>.
- Johnson, T. A., Vervoort, J. D., Ramsey, M. J., Aleinikoff, J. N., Southworth, S. (2018). Constraints on the timing and duration of orogenic events by combined Lu–Hf and Sm–Nd geochronology: An example from the Grenville orogeny. *Earth Planet. Sci. Letters*, v. 501, p. 152–164.

Literature that provides examples of the application of Sm-Nd isotope, Pb-Pb, and Rb-Sr geochemistry to ultramafic cumulates and magmas associated with copper porphyry deposits:

- Bouse, Robin M., Joaquin Ruiz, Spencer R. Titley, Richard M. Tosdal, and Joseph L. Wooden. "Lead Isotope Compositions of Late Cretaceous and Early Tertiary Igneous Rocks and Sulfide Minerals in Arizona; Implications for the Sources of Plutons and Metals in Porphyry Copper Deposits." *Economic Geology* 94, no. 2 (1999): 211–44.
<https://doi.org/10.2113/gsecongeo.94.2.211>.
- Lang, James R., and Spencer R. Titley. "Isotopic and Geochemical Characteristics of Laramide Magmatic Systems in Arizona and Implications for the Genesis of Porphyry Copper Deposits." *Economic Geology* 93, no. 2 (1998): 138–70.
<https://doi.org/10.2113/gsecongeo.93.2.138>.

D. Scope

We have collected samples from a variety of Laramide igneous rocks and ultramafic xenoliths for Sm-Nd isotopic analysis. In combination with major and trace element geochemistry, the Sm-Nd data will be used constrain magmatic source(s) and petrogenetic models.

II. WORK REQUIREMENTS

A. Technical Requirements

1. Work Requirements/Products

The USGS will provide 25 surface whole rock samples for the Sm-Nd, U-Pb, and Rb-Sr analyses. The samples have been pulverized in agate or alumina corundum (avoiding metal) and at least 1 g of powder per sample is available to submit to the lab along with garnet separates for the few samples that contain the mineral. The submitted samples will be analyzed using recognized spikes, separation methods, standards, and instrumental techniques to meet the typical precision for analyses reported throughout the literature ($2\sigma < \pm \sim 0.00001$).

2. Criteria for acceptance of Product

Analyses will be accepted if they meet the requirements of the lab's quality control and quality assurance specialists.

B. Deliverables

1. Characteristics for work product described in II.A.1.

Deliverables will include Sm-Nd, Rb-Sr, and U-Pb elemental and isotopic compositions of the 25 samples submitted. A report will be submitted to the USGS upon completion of the work. This report will contain all relevant isotopic data that would routinely be included in scientific papers. This includes:

- A description of analytical techniques and methods of data reduction
- Table of the isotopic data, including elemental and isotopic concentrations of Sm, Nd, U, Pb, Rb, and Sr along with their uncertainties.

The contractor will communicate with the USGS on a regular basis and in case of any unforeseen complications with the rock samples submitted.

2. Delivery dates

Depending on lab schedules, the work for 25 samples should be completed within 12 months from the award of the contract. Digital data can be transferred to the USGS by email or other means of data delivery. Excess powdered sample should be returned to the USGS.

III. SUPPORTING INFORMATION

A. Place of Performance

At the contractor's lab.

B. Period of Performance

The work will require ~30 days of lab time, performed within 12 months from the award of the contract.

C. Government Furnished Property (GFP): None.

D. Special Considerations

1. Key personnel qualifications:

The lead isotope geochemistry expert of the lab should have a Ph.D degree and have 5 or more additional years of experience. They should have authorship on several peer-reviewed publications that report Sm-Nd, Rb-Sr, and U-Pb analyses for igneous rocks. The lab must be an internationally recognized and reputable isotope geochemistry lab.

2. Data Rights, Safety, Environment: None

3. Security/Privacy: None